WORKED SOLUTION

STEP II 2003, Question 1

Pure • Simultaneous Linear Equations

STEP II 2003, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q1

The key idea

The unknown x only ever appears as ax, and y appears with coefficient -1 in every equation. So subtracting consecutive equations kills y completely, and the two differences we get have the same left-hand side. That makes the system consistent only for one value of b, and once it is consistent there is one free parameter. The rest of the question is about choosing that parameter well.

Solution

Number the equations (1), (2), (3). Subtracting (1) from (2), and (2) from (3), gives

$$ax - 2z = 4, \tag{A}$$
$$ax - 2z = b - 7. \tag{B}$$

Any solution of the system satisfies both (A) and (B), so a solution can exist only if $4 = b - 7$, that is $b = 11$. This holds for every value of a.

Part (i)

With $a = 0$ the equations become $-y - z = 3$, $-y - 3z = 7$, $-y - 5z = b$, and x does not appear at all.

If there is a solution, then $b = 11$ by the remark above.

Conversely, suppose $b = 11$. Equation (A) now reads $-2z = 4$, so $z = -2$, and then (1) gives $y = -3 - z = -1$. Checking all three equations with $y = -1$, $z = -2$:

$$1 + 2 = 3, \qquad 1 + 6 = 7, \qquad 1 + 10 = 11 = b,$$

so (with any x at all) we have a solution. Hence the equations have a solution if and only if $b = 11$.

Part (ii)

Now $a \neq 0$ and $b = 11$. Put $z = \lambda$. Equation (A) gives $ax = 4 + 2\lambda$, and since $a \neq 0$ we can divide:

$$x = \frac{4 + 2\lambda}{a}.$$

Then (1) gives $y = ax - z - 3 = 4 + 2\lambda - \lambda - 3 = 1 + \lambda$.

We found these values using only (1) and (2), so we must check they satisfy (3) too:

$$3ax - y - 5z = 3(4 + 2\lambda) - (1 + \lambda) - 5\lambda = 12 + 6\lambda - 1 - 6\lambda = 11 = b.$$

So for every number λ,

$$x = \frac{4 + 2\lambda}{a}, \qquad y = 1 + \lambda, \qquad z = \lambda$$

is a solution with $z = \lambda$, as required. (In fact these are *all* the solutions: any solution has some value λ of z, and the working above then forces x and y.)

Ask yourself Why do we have to check equation (3) separately?

We used (1) and (2) (in the form (1) and (A)) to find x and y. Three equations in three unknowns do not automatically fit together, and (3) is exactly where b comes in. The check is where $b = 11$ gets used.

Part (iii)

With $a = 2$, part (ii) gives every solution as $(x, y, z) = (2 + \lambda,\ 1 + \lambda,\ \lambda)$. Then

$$\begin{aligned} x^2 + y^2 + z^2 &= (2 + \lambda)^2 + (1 + \lambda)^2 + \lambda^2 \\ &= 3\lambda^2 + 6\lambda + 5 \\ &= 3(\lambda + 1)^2 + 2, \end{aligned}$$

which is least, with value 2, exactly when $\lambda = -1$. So the required solution is

$$x = 1, \qquad y = 0, \qquad z = -1.$$

Geometrically, the solutions form a line with direction $(1, 1, 1)$, and we are finding the point of the line closest to the origin. Indeed $(1, 0, -1) \cdot (1, 1, 1) = 0$, so the position vector is perpendicular to the line, as it should be.

Part (iv)

By the remark at the start, b must be 11. If $a = 0$, part (i) forces $y = -1$, $z = -2$, so $y^2 + z^2 = 5$, which is too big. So $a \neq 0$, and by part (ii) the solutions are $x = (4 + 2\lambda)/a$, $y = 1 + \lambda$, $z = \lambda$.

Ask yourself How can x be huge while y and z stay small?

The only place a appears is in $x = (4+2\lambda)/a$. Choosing λ fixes y and z, and then a small enough a makes x as large as we like.

We need $y^2+z^2 = (1+\lambda)^2+\lambda^2 = 2\lambda^2+2\lambda+1 < 1$, that is $2\lambda(\lambda+1) < 0$, that is $-1 < \lambda < 0$. Take $\lambda = -\frac{1}{2}$: then $y = \frac{1}{2}$, $z = -\frac{1}{2}$ and $y^2+z^2 = \frac{1}{2} < 1$, while $x = 3/a$. Choosing

$$a = 10^{-6}$$

gives $x = 3\times 10^6 > 10^6$. So $a = 10^{-6}$ (with $b = 11$) works.

More generally, for $-1 < \lambda < 0$ we have $2 < 4+2\lambda < 4$, so a suitable solution exists exactly when $0 < a < 4\times 10^{-6}$. Negative a cannot work, since then $x < 0$.

Checks

- Substitute $a = 2$, $(x,y,z) = (1,0,-1)$ into the original equations: $2-0+1 = 3$, $4-0+3 = 7$, $6-0+5 = 11$.
- For part (iv), with $a = 10^{-6}$ and $(x,y,z) = (3\times 10^6, \frac{1}{2}, -\frac{1}{2})$: $3-\frac{1}{2}+\frac{1}{2} = 3$, $6-\frac{1}{2}+\frac{3}{2} = 7$, $9-\frac{1}{2}+\frac{5}{2} = 11$.
- We checked with exact fractions that the part (ii) formula satisfies all three equations for hundreds of random a and λ, and a grid search over λ confirms the minimum value 2 at $\lambda = -1$ in part (iii).

Common slips

- Finding x and y from two of the equations and forgetting to check the third.
- In part (iv), dismissing it as impossible because y and z "can't both be small". They can: $y - z = 1$ only forces $y^2+z^2 \geq \frac{1}{2}$.

Beyond STEP

This is a first look at rank and consistency in linear algebra. The coefficient matrix here is singular (its rows satisfy $r_1 - 2r_2 + r_3 = 0$), so the system has either no solutions or a whole line of them, depending on whether the right-hand side satisfies the same relation. Part (iv) also shows why nearly singular systems are dangerous in numerical work: a tiny coefficient a produces an enormous solution. Strang's *Introduction to Linear Algebra* covers all of this, including least-norm solutions like the one in part (iii).

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